Four Labels, Two Questions

CS383 · Lecture 9 unplugged activity · work in pairs · about 15 minutes

A classifier makes four kinds of outcome. **TP, TN, FP, FN** are just names for them. You never need to memorize which is which — you can work each one out with two questions.

Question 1

What did the **MODEL** say? Ignore reality for now.

It said the thing we are hunting for → **Positive**
It said the other thing → **Negative**
This gives you the **second word**.

Question 2

Was the model **RIGHT**? Compare it to reality.

Model matches reality → **True**
Model does not match → **False**
This gives you the **first word**.

Example: the smoke alarm sounds, but it was only shower steam. Q1: it said “fire” → **Positive**. Q2: there was no fire, so it was wrong → **False**. Put them together: **False Positive (FP)**.

“Positive” does not mean “good.” The positive class is simply the thing the model is hunting for: spam, an illness, a fire, fraud. Each card tells you what the positive class is.

What you will do

Round 1 (6 min). Cut out the 16 cards. For each card, answer the two questions and write TP, TN, FP or FN on the worksheet (page 3).

Round 2 (3 min). Place every card on the 2×2 mat (page 2). Rows = reality, columns = what the model said. Compare with your partner.

Round 3 (3 min). For each scenario, decide which mistake (FP or FN) is worse, and why (page 4).

Round 4 (2 min). Flip the positive class and see what happens to the labels (page 4).

The 2 × 2 Mat

Place each card in the cell where it belongs. Rows = what is TRUE. Columns = what the MODEL said.

What is TRUE (reality)	What the MODEL said: POSITIVE	What the MODEL said: NEGATIVE
POSITIVE	Label this cell: _____	Label this cell: _____
NEGATIVE	Label this cell: _____	Label this cell: _____

Check yourselves: each cell should hold 4 cards, one from each scenario. The two cells where the model was **right** (True) sit on one diagonal; the two **mistakes** (False) sit on the other.

Round 1 Worksheet

For each card: answer the two questions, then write the label.

Card	Q1: Model said…	Q2: Model right?	Label
A	Positive / Negative	Yes / No	_____ (TP TN FP FN)
B	Positive / Negative	Yes / No	_____ (TP TN FP FN)
C	Positive / Negative	Yes / No	_____ (TP TN FP FN)
D	Positive / Negative	Yes / No	_____ (TP TN FP FN)
E	Positive / Negative	Yes / No	_____ (TP TN FP FN)
F	Positive / Negative	Yes / No	_____ (TP TN FP FN)
G	Positive / Negative	Yes / No	_____ (TP TN FP FN)
H	Positive / Negative	Yes / No	_____ (TP TN FP FN)
I	Positive / Negative	Yes / No	_____ (TP TN FP FN)
J	Positive / Negative	Yes / No	_____ (TP TN FP FN)
K	Positive / Negative	Yes / No	_____ (TP TN FP FN)
L	Positive / Negative	Yes / No	_____ (TP TN FP FN)
M	Positive / Negative	Yes / No	_____ (TP TN FP FN)
N	Positive / Negative	Yes / No	_____ (TP TN FP FN)
O	Positive / Negative	Yes / No	_____ (TP TN FP FN)
P	Positive / Negative	Yes / No	_____ (TP TN FP FN)

Circle your choice in Q1 and Q2. Tip: Q1 gives the second word; Q2 gives the first word.

Rounds 3 and 4

Think, discuss with your partner, then write.

Round 3 — Which mistake is worse?

A **false positive (FP)** is a false alarm. A **false negative (FN)** is a miss. In each scenario, circle the mistake you think is worse and give a reason in one short sentence.

Spam filter (positive = spam)
Worse mistake: FP / FN Because: _____

_____

Strep throat test (positive = has strep throat)
Worse mistake: FP / FN Because: _____

_____

Smoke alarm (positive = a fire)
Worse mistake: FP / FN Because: _____

_____

Card fraud detector (positive = fraud)
Worse mistake: FP / FN Because: _____

_____

Round 4 — Flip the positive class

The spam team changes its mind: from now on **"NOT spam" is the positive class** (so the filter is hunting for good email). Nothing about the emails or the filter changes. Re-label the four spam cards.

Card	What happened	Old label	New label
F	"WIN a free cruise — click now!"	**TP**	_____
L	a coworker asking about lunch	**TN**	_____
N	"Your bank account is locked — log in here"	**FN**	_____
O	your professor's assignment reminder	**FP**	_____

What pattern do you see? When we flip which class is positive, TP and TN _____, and FP and FN _____.

A **Smoke alarm**
Hunting for (POSITIVE): a fire

THE CASE
Grease fire on the stove

THE MODEL SAID
ALARM SOUNDS

IN REALITY
There is a fire

B **Smoke alarm**
Hunting for (POSITIVE): a fire

THE CASE
Quiet night at home

THE MODEL SAID
ALARM STAYS SILENT

IN REALITY
There is no fire

C **Card fraud detector**
Hunting for (POSITIVE): fraud

THE CASE
Stolen card used to buy gift cards

THE MODEL SAID
APPROVED

IN REALITY
It is fraud

D **Card fraud detector**
Hunting for (POSITIVE): fraud

THE CASE
Cardholder's own plane ticket, bought abroad

THE MODEL SAID
FLAGGED as fraud

IN REALITY
It is legitimate

E **Smoke alarm**
Hunting for (POSITIVE): a fire

THE CASE
Fire smoldering inside a wall

THE MODEL SAID
ALARM STAYS SILENT

IN REALITY
There is a fire

F **Spam filter**
Hunting for (POSITIVE): spam

THE CASE
Email: "WIN a free cruise — click now!"

THE MODEL SAID
SPAM

IN REALITY
It really is spam

G **Strep throat test**
Hunting for (POSITIVE): has strep throat

THE CASE
Healthy patient with a mild cold

THE MODEL SAID
Test result: POSITIVE

IN REALITY
The patient has no strep

H **Card fraud detector**
Hunting for (POSITIVE): fraud

THE CASE
$37 weekly grocery run

THE MODEL SAID
APPROVED

IN REALITY
It is legitimate

I **Strep throat test**
Hunting for (POSITIVE): has strep throat

THE CASE
Patient with a sore throat who has strep

THE MODEL SAID
Test result: POSITIVE

IN REALITY
The patient has strep

J **Strep throat test**
Hunting for (POSITIVE): has strep throat

THE CASE
Healthy patient, routine visit

THE MODEL SAID
Test result: NEGATIVE

IN REALITY
The patient has no strep

K **Smoke alarm**
Hunting for (POSITIVE): a fire

THE CASE
Steam from a hot shower

THE MODEL SAID
ALARM SOUNDS

IN REALITY
There is no fire

L **Spam filter**
Hunting for (POSITIVE): spam

THE CASE
Email: a coworker asking about lunch

THE MODEL SAID
NOT SPAM

IN REALITY
It is not spam

M **Strep throat test**
Hunting for (POSITIVE): has strep throat

THE CASE
Patient with a sore throat who has strep

THE MODEL SAID
Test result: NEGATIVE

IN REALITY
The patient has strep

N **Spam filter**
Hunting for (POSITIVE): spam

THE CASE
Email: “Your bank account is locked — log in here”

THE MODEL SAID
NOT SPAM

IN REALITY
It really is spam (phishing)

O **Spam filter**
Hunting for (POSITIVE): spam

THE CASE
Email: your professor’s assignment reminder

THE MODEL SAID
SPAM

IN REALITY
It is not spam

P **Card fraud detector**
Hunting for (POSITIVE): fraud

THE CASE
$2,400 electronics order on a stolen card

THE MODEL SAID
FLAGGED as fraud

IN REALITY
It is fraud